LAB NOTEBOOK ▪ 2026-10-22 ▪ 7 min read

Evolving a Float Kernel, Then Checking It

AlphaEvolve wrote 19 versions of a WENO5 kernel in float. Three beat the seed. Verification showed why it helps, and found the event where no float kernel can help.

1.31×	2.73×	19/19	$5.58
lower error, scored window	before $t = 1.45$	checks pass	in agent tokens

The ask

The last session (`report.tex`) ended with the optimize stage planned, calibrated and red-teamed, and blocked on Google Cloud credentials. The scientist came back with:

> *Let continue the experiment I have GCP credentials.*

They also pasted the notebook settings they use elsewhere: project, engine, model, job size.

Getting to launch

The credentials weren't where the container looks. The agent checked `~/.config/gcloud` (no ADC file, no host gcloud) before trying anything.

"How should I login?" Through the container itself, so the credentials land in the mounted folder. **"Write that code to file so I can run."** A three-line script went to `/tmp`, outside the repo. One login later the container had a token.

Three small mismatches followed, each fixed from the tool's own source, not guessed: `ae` wants the numeric project number, not the name; it rejects the notebook's model preset `GEMINI_V3P0_MIXTURE`, which in `ae` is just `gemini-3-flash-preview` at weight 1; and the launch script crashed because `experiment start` does not print JSON.

"Make the changes you need, and be careful with credentials." The agent fixed the script, then searched the repo and logs for tokens, refresh tokens and keys (none), and confirmed `results/ae/` is gitignored and the ADC file is mode 600.

The run that scored nothing

The first experiment, `exp-free-hyena`, finished in six minutes: 9 candidates, 9 nulls, "invalid score". Bad candidates, or a bad evaluator? The agent pulled one and scored it by hand: 9.20, valid.

The bug was ours. `ae` copies `evaluator.py` into a fresh temporary folder and runs it there, so the output path the evaluator derived from its own location pointed into that

folder, whose `results/` did not exist. The app creates only the last folder, and crashed. The red team had run the evaluator in place, never the way `ae` runs it. One line in `evolve.py` (create the parent first) fixed it for every future experiment, and the rerun scored 9 of 10.

What evolution found

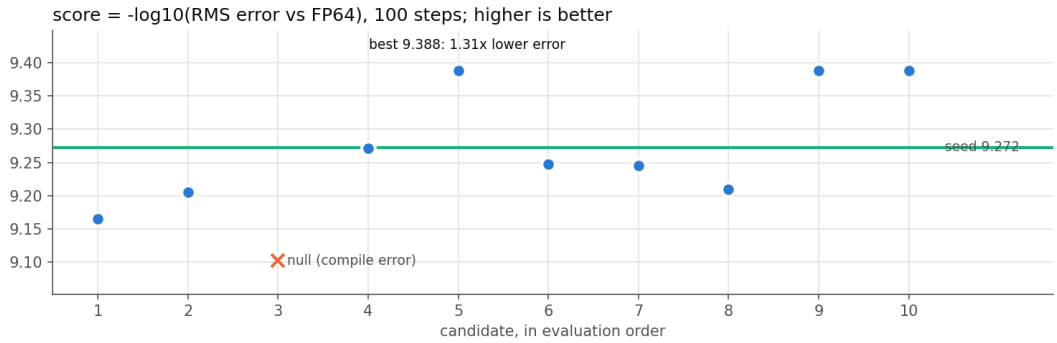


Figure 1: The second run, `exp-ginger-chimpanzee`: 10 candidates from Gemini 3 Flash, each built, audited (no FP64 in the block, nothing changed outside it) and run in the container.

Three candidates tied near 9.388. They share one idea. The seed computes a WENO5 face value as a weighted sum of three stencils, each of size $|\phi|$, whose large parts cancel:

$$f = a_1 f_1 + a_2 f_2 + a_3 f_3.$$

The candidates eliminate $a_2 = 1 - a_1 - a_3$ and write it as the centre value plus a small correction built from neighbour differences $v_k = s_k - s_{k+1}$:

$$f = s_3 + \tfrac{1}{6}(2a_1(v_1 - 2v_2 + v_3) + a_3(v_2 - 2v_3 + v_4) - v_2 - 2v_3).$$

In a smooth field the differences are $O(\Delta x)$, so float rounds the small part, not the big one. A textbook trick, found unprompted. Running all three to $t = 2$ picked `humongous-macaw`: tied at 100 steps, best over the full run.

Verification

run	seed error	candidate	gain
eval, 128^2, 100 steps	5.34×10^{-10}	4.09×10^{-10}	1.31×
256^2, 200 steps	5.54×10^{-10}	4.36×10^{-10}	1.27×
to $t = 2$, before the jump	9.12×10^{-9}	3.34×10^{-9}	2.73×
to $t = 2$, after both jumps	4.87×10^{-7}	5.27×10^{-7}	0.92×

All 19 checks pass. The field is bit-identical on 1, 2 and 4 ranks. Area and mass at $t = 2$ stay at fp32 level (2×10^{-6} off FP64; bf16 is 3×10^{-3}). And sympy confirms the

candidate's face value equals the seed's in exact arithmetic: the same scheme, rounded better.

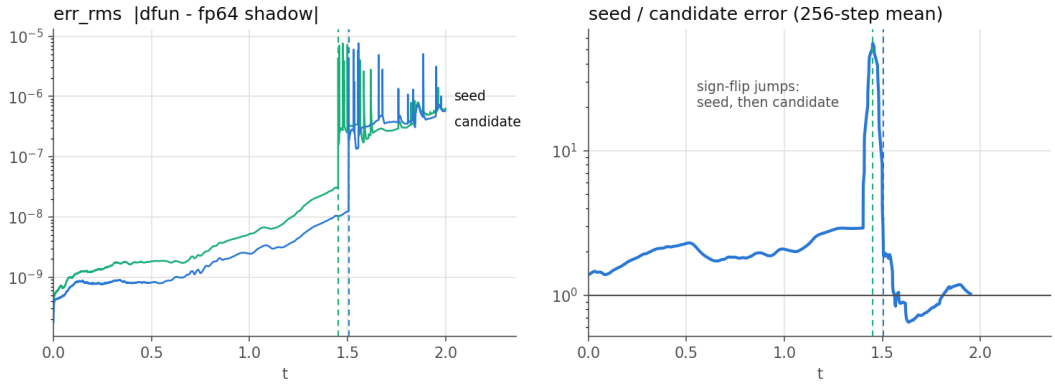


Figure 2: Error vs the FP64 shadow over the full run. The candidate stays 2–3× below the seed until a jump at $t \approx 1.45$ (seed) or 1.51 (candidate); after it the two are alike.

The jump. In one step, the seed's largest error goes from 1.7×10^{-7} to 5.5×10^{-4}. Rerunning to the step before and after found one cell crossing zero: $\phi = +1.18 \times 10^{-8}$ in FP64, -3.25×10^{-9} in float. The FP64 redistance tests for interface cells by sign (`redistance.cpp:70`), takes the other branch, and the neighbour moves by 5.5×10^{-4}: a 37 000× amplification. Every low-precision run hits such switches; better rounding only delays the first one by 275 steps. After that, sign flips decide the error, not the kernel.

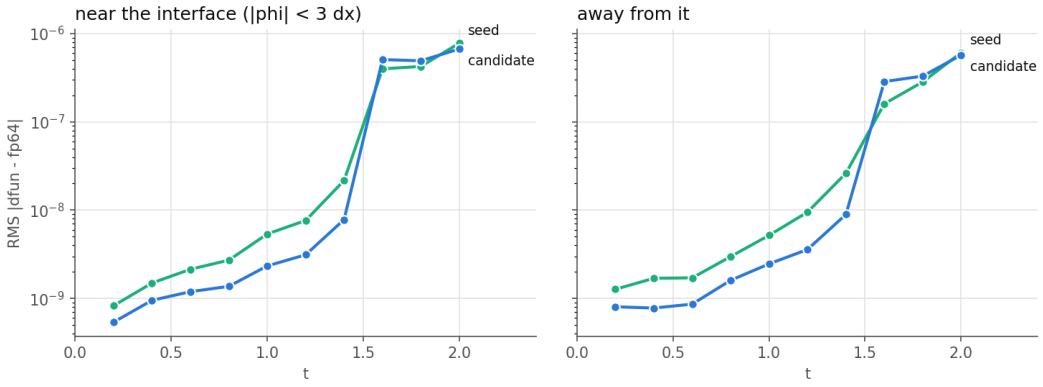


Figure 3: The gain is the same near the interface and away from it, until the jump.

Verdict (the scientist decides). Merge: same scheme, more accurate while the flow is smooth, no worse at $t = 2$ overall. A further optimize run should score a window or a quantity the sign switch cannot decide: $t < 1.4$, $2\times$ resolution, or area and mass.

How it went

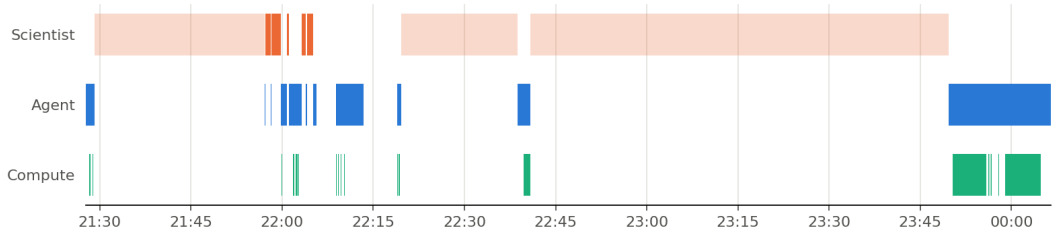


Figure 4: Pale orange: the scientist was away, mostly while AlphaEvolve ran on Google Cloud (not on the compute lane, which counts local container runs only).

Nine messages, 120 words from the scientist. The longest one was *"lets run verification on these results and prepare a comprehensive report"*; the agent took that as approval of Gate 2 and the verify plan, chose the defaults, and recorded each as its own choice in the plans.

What it cost

Claude Opus 5.5, 108 calls	tokens	$/M	$
fresh input	218	4.00	0.00
cache writes	519 745	5.00	2.60
cache reads	8 416 338	0.20	1.68
output	64 619	20.00	1.29
total			**5.58**

AlphaEvolve's side, 19 candidates over both runs: 58 845 input and 612 697 output tokens of Gemini 3 Flash, billed to the GCP project (not priced here). Local compute: 16 minutes of container runs, all of verification included.

Human time vs. time saved

Doing it by hand (our estimate, for an expert new to ae)	
credentials, project number, model names, launch	1 h
debug 9 nulls to the evaluator's working directory	1 h
re-score and full-run the top candidates, pick one	1 h
verification script, runs, the sign-flip hunt	3 h
write this up with figures and costs	2 h
total	**≈ 8 h**
With the agent: scientist's attention	**5 min**

The agent worked 32 of the 162 minutes; most of the rest was waiting on Google Cloud.

What we'd change

- **Red-team the way `ae` runs:** copy `evaluator.py` to a temporary folder first. It would have saved a whole run.
- **Score past the easy window:** 100 steps hid the jump; 2.73× early, 1.17× overall.
- **Map notebook settings to `ae`:** presets like `GEMINI_V3P0_MIXTURE` and project names need translating; the stage could do it.
- **Still a papercut:** the app should create its output folder's parents.

Numbers: `3-verify/results/verify/summary.md`, `3-verify/analysis.md`, `2-optimize/results/scores.csv`; cost and timeline from `session.py` on this session's transcript. Plans and patches are under `experiments/2026-10-20/`.